

HAT-P-36b

R-0704

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_260327 · created 2026-07-27T22:04:36+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461126.7748 +/- 0.0065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.091 +/- 0.015
Transit depth (Rp/Rs)^2	0.83 +/- 0.27 [%]
Orbital Inclination (inc)	89.67 +/- 2.86
Airmass coefficient 1 (a1)	1.114 +/- 0.013
Airmass coefficient 2 (a2)	-0.103 +/- 0.038
Scatter in the residuals of the lightcurve fit is	1.14 %
Best Comparison Star	#2 - [229, 452]
Optimal Aperture	3.6
Optimal Annulus	14.36
Transit Duration (day)	0.0925 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.091 $\pm$ 0.015 vs 0.126 (deviation 1.97 σ)
Duration fit vs published	0.0925 d vs 0.0925 d (deviation 0.0 σ)
Mid-time O–C	13.9 min
Depth SNR	5.1
Residual scatter	1.14 %

Light curve

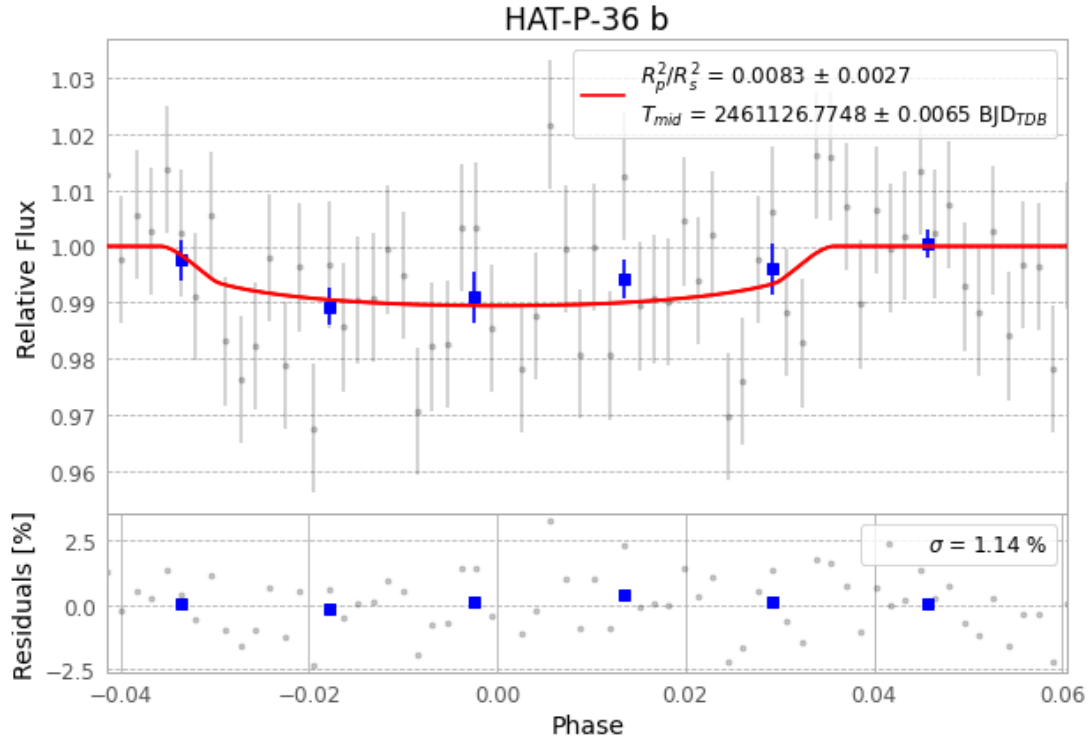


Figure 1 — FinalLightCurve_HAT-P-36 b_2026-03-26.png (SHA-256 28b75f3a6af062a4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [296, 195], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a5e194ea6d1267d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

66 FITS frames (43 MB), HATP-36260327050915.FITS → HATP-36260327082415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-260327.log (9e0eb307ec42...), FinalLightCurve_HAT-P-36 b_2026-03-26.png (28b75f3a6af0...), FinalLightCurve_HAT-P-36 b_2026-03-26.csv (4da9444cf54a...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 22:04Z.